



MME 321 Final: Elevator Pulley System

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1.0 Derivation of Load Torque

The load torque is essential to the motion of the elevator, as it will only be used when the elevator is carrying a load (people). The load torque can be analytically modeled as follows:

The Tension on the elevator wire can be modeled as

$$T_1 = m_{\text{cart}} * y'' + b * y' + m_{\text{cart}} * g$$

The tension on the counterweight wire can be modeled as

$$T_2 = m_{\text{counter}} * g - m_{\text{counter}} * y''$$

Using the radius of the pulley, the torque acting on the pulley can then be calculated as

$$T_p = J_p \Theta'' + T_b + T_1 * r - T_2 * r$$

With proper substitution, the equation for Load Torque becomes,

$$T_L = (J_p * N + (m_{\text{cart}} + m_{\text{counter}}) * r^2 * N) * \Theta_m'' + (b_{\text{cart}} * r^2 * N + B_p * N) * \Theta_m' + (m_{\text{cart}} - m_{\text{counter}}) * r * g$$

The Transfer functions used in this experiment are derived in previous labs and take the form:

$$\frac{I(s)}{V(s)} = \frac{1}{(La*s + Ra)}$$

Which is used to convert input voltage to current. And,

$$\frac{\omega(s)}{T(s)} = \frac{1}{(J_{\text{eff}} * s + b_{\text{eff}})}$$

Which is used to convert Torque to angular velocity.

J_{eff} can be calculated by adding the load inertia and the motor inertia.

B_{eff} can be calculated by adding the load coefficient of friction and the motor coefficient of friction.

The following table outlines all of the parameters that were used to model this system within simulink, matlab, and simscape.

Elevator Cart Parameters

Parameters	symbol	Value
Cart mass	m_{cart}	1360kg
Travel height	h_{max}	10m

Friction coefficient	b	50 N*s/m
Counter mass	m_{counter}	1000kg
Cable tension	T	varies

DC Motor Parameters

Parameters	symbol	Value
Rated Voltage	V	100V
Armature Resistance	R_a	2.5 Ω
Armature Inductance	L_a	0.05 H
Torque Constant	K_t	0.8 N*m/A
Back EMF const	K_e	0.8 V*s/rad
Rotor Inertia	J_m	kg*m ²
Motor Friction	B_m	N*m*s/rad

Pulley and Wire Parameters

Parameters	symbol	Value
Pulley Inertia	J_p	0.085 kg*m ²
Pulley Friction	B_p	0.001
Pulley Radius	r	0.2m
Gear Ratio	N	10:1

2.0 Simulink and Simscape Diagrams

The Elevator pulley system was modeled in both simulink and simscape to test responses of the elevator's speed and position when moving floors. The following diagrams were constructed and simulated.

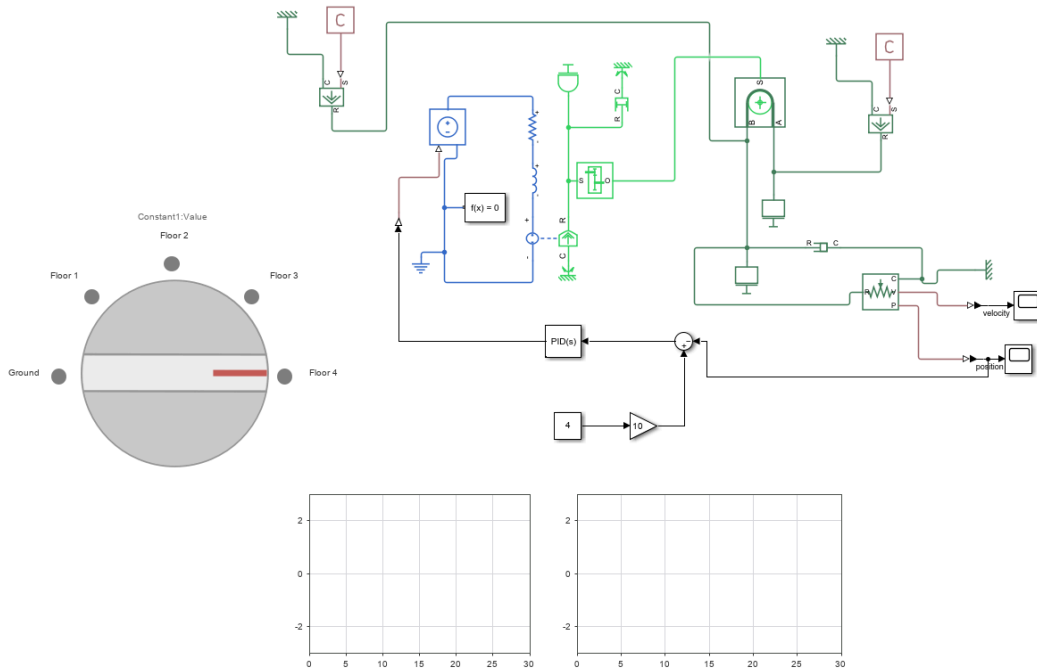


Fig. 1 Simscape Elevator Simulation Schematic

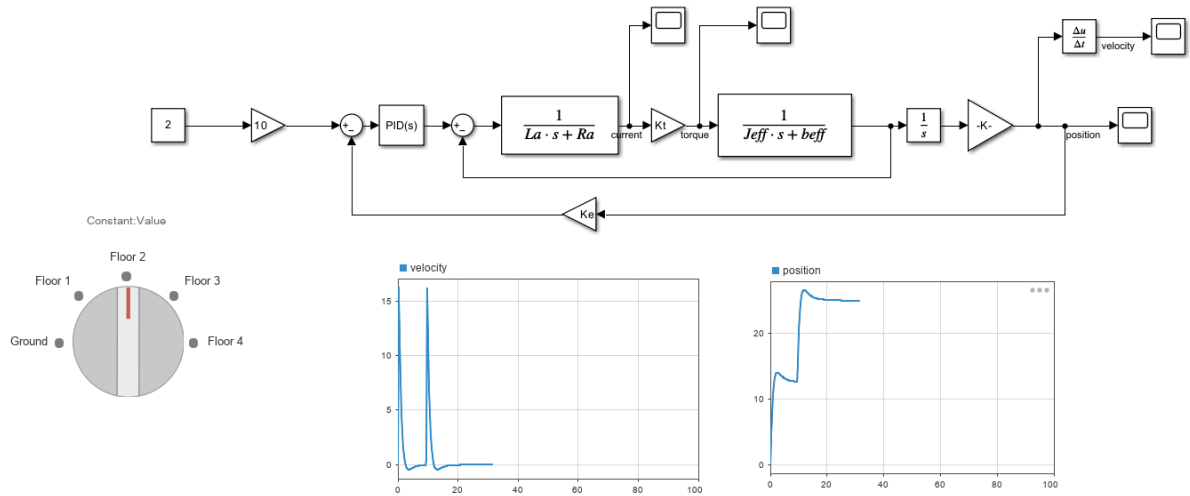


Fig. 2 Simulink Elevator Simulation Schematic

3.0 Simulation Responses

For the purposes of this experiment, the position and velocity of the elevator were recorded and graphed on both simscape and simulink models. Additionally, the armature current and torque of the motor were plotted on simulink.

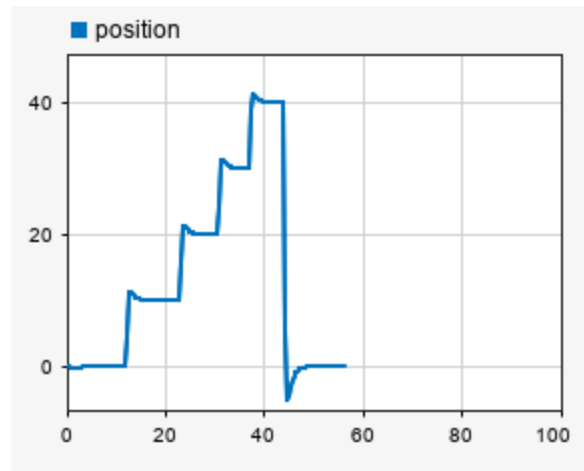


Fig. 3 Simscape elevator position response

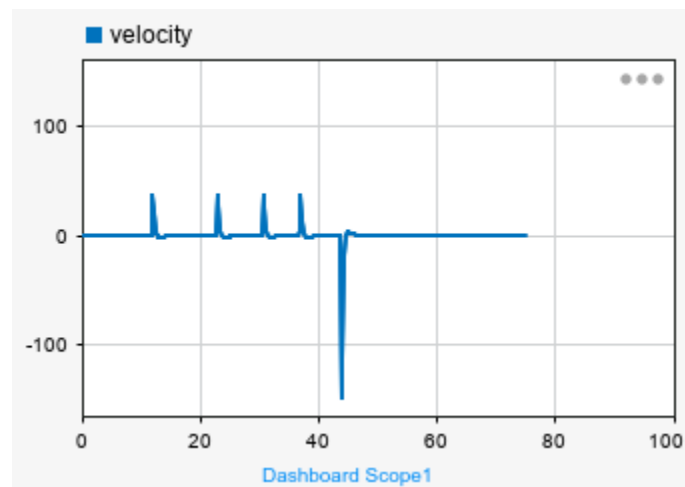


Fig. 4 Simscape elevator velocity response

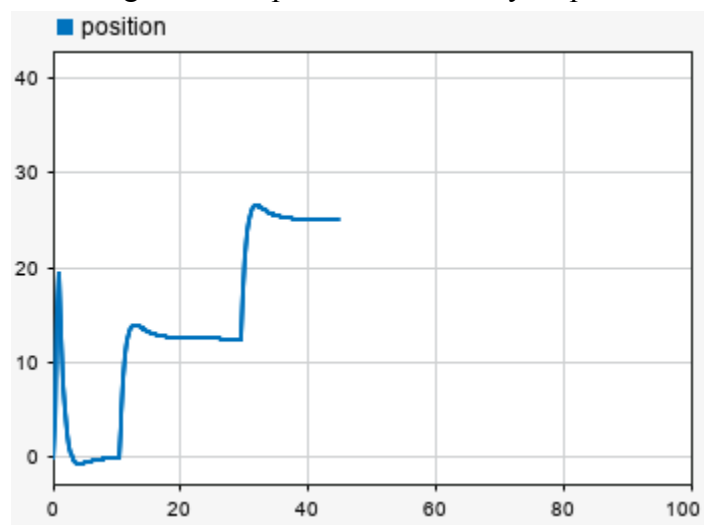


Fig. 5 Simulink elevator position response

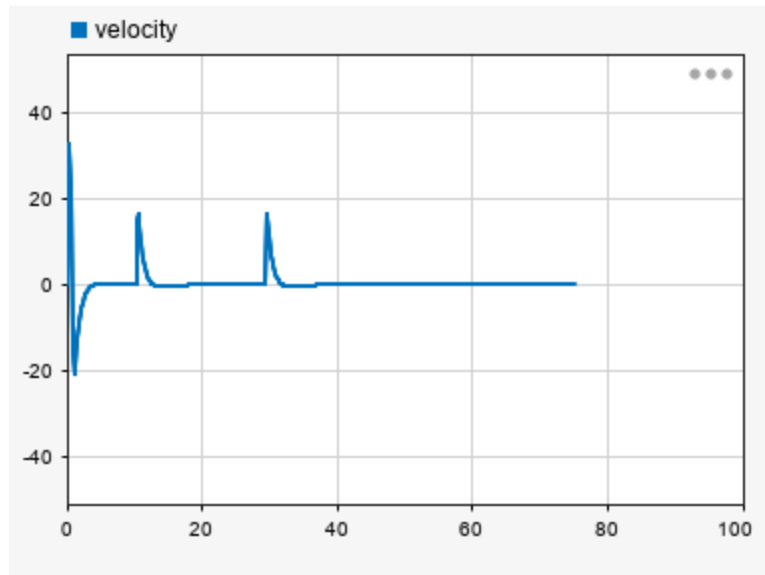


Fig. 6 Simulink elevator velocity response

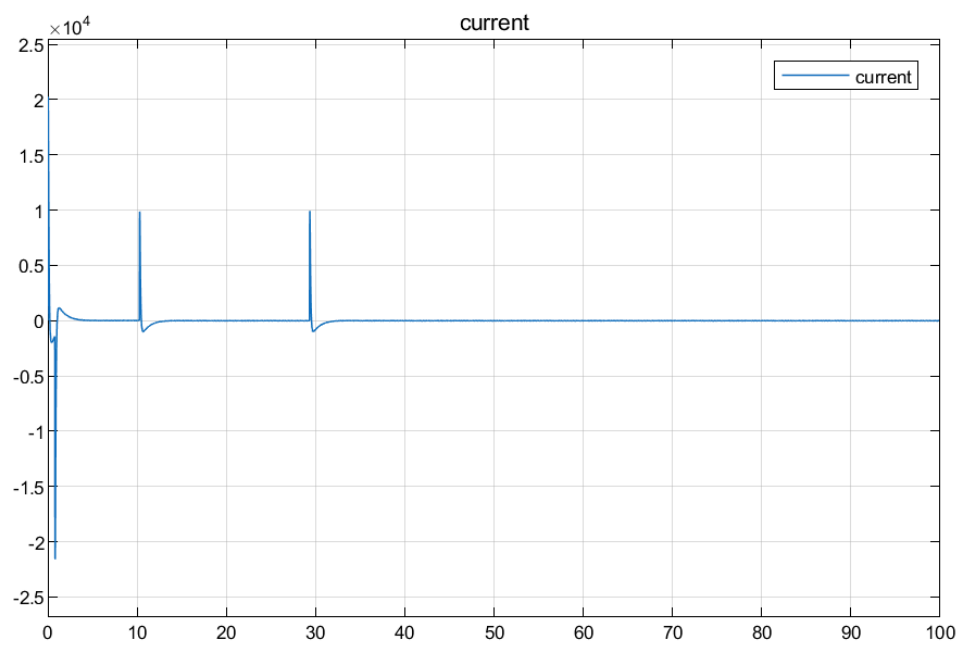


Fig. 7 Simulink armature current response

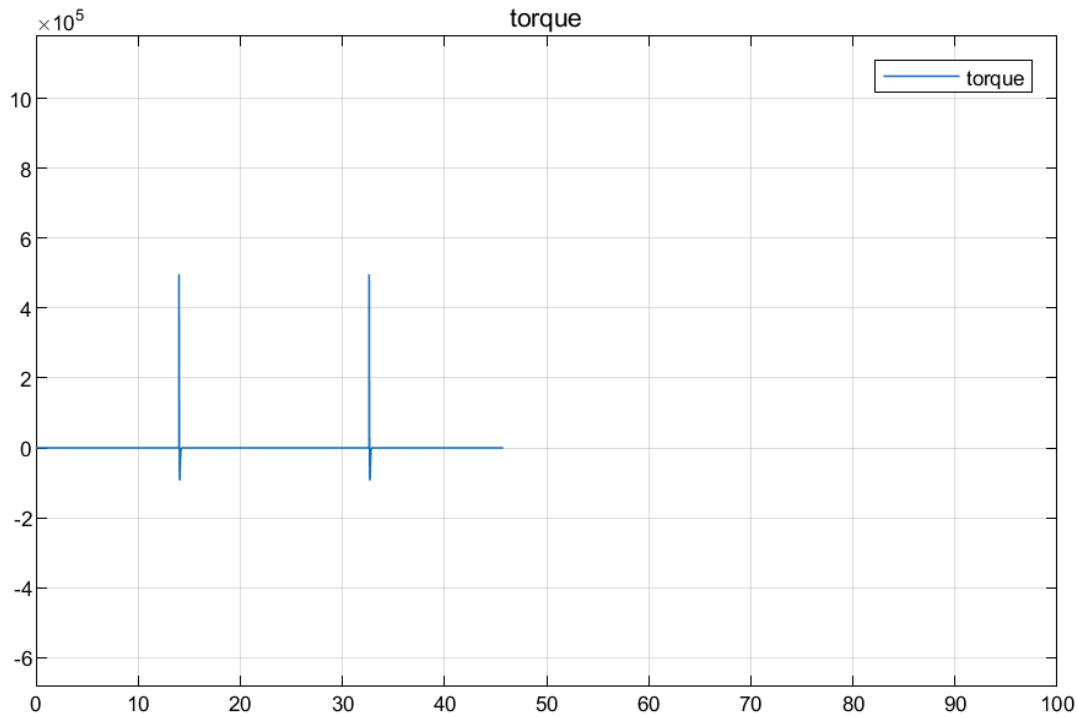


Fig. 8 Simulink Motor Torque response

8.0 PID Tuning

To achieve stable and accurate position control of the elevator system, a PID controller was implemented in the Simulink model. The PID controller works by continuously computing the error between the desired floor position and the actual position of the elevator cart, then adjusting the motor input voltage accordingly through proportional, integral, and derivative action.

The gain was mainly controlled through the tuning window, where two slider bars control the response time and transient behavior. The PID block then automatically updates its K_p , K_d , and K_i values.

The proportional gain K_p directly drives the system response toward the setpoint. A higher K_p reduced rise time but introduced overshoot. The derivative gain K_d was then introduced to dampen the oscillatory behavior observed in both the velocity and position responses such as Figs. 5 and 6, particularly the sharp velocity spikes seen during floor transitions. Finally, the integral gain K_i was adjusted to eliminate steady-state error, ensuring the elevator cart reaches and holds the target floor position without a persistent offset.

The transient spikes in Fig. 7 and 8 occur at each floor transition and are likely facing some sort of parameter issue. This makes sense because the steady state settling value for the simulink diagram is incorrect for a varying amount at each floor.

9.0 Conclusion

This project successfully modeled an elevator pulley system driven by a DC motor using Simscape, and almost correctly using simulink. The transfer functions that were generated in MATLAB were implemented within simulink to simulate realistic elevator dynamics across multiple floors. Both simulations produced position and velocity responses, validating the mathematical model. The Simscape model offered a more physically intuitive representation, while the Simulink model allowed visualizing armature current and motor torque. The PID controller provided closed-loop position control.